MIL-W-20629
28 March 1952

MILITARY SPECIFICATION

WATCHES, WRIST (WATERTIGHT)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Navy 18W8 (SHIPS)
15 January 1947

This specification consists of this cover sheet and Navy Department Specification 18W8(SHIPS) dated 15 January 1947, together with Amendment-1 dated 15 February 1948, attached hereto, without modification.

Copies of this specification may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, New York. Both the title and identifying number or symbol should be stipulated when requesting copies.

When a request for this specification is received by a supplying activity, it will be necessary to attach this cover sheet to the pertinent specification before issue.

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18W8(SHIPS)

January 15, 1947
SUPERSEDING
18W8(INT)
December 1, 1944

BUREAU OF SHIPS SPECIFICATION

WATCHES, WRIST (WATERTIGHT)

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A-1. Specifications.- The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.

JOINT ARMY-NAVY SPECIFICATIONS

JAN-P-105 - Boxes, Wood, Cleated, Plywood.
JAN-P-106 - Boxes, Wood, Nailed.
JAN-P-116 - Preservation Methods of.
JAN-P-125 - Barrier-Materials, Waterproof, Flexible.
JAN-P-140 - Adhesives, Water-Resistant, Case Liner.

FEDERAL SPECIFICATION

TT-R-58 - Radioactive-Luminous-Compound and Adhesives.

A-2. Other publications.- The following Bureau of Supplies and Accounts publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Navy Shipment Marking Handbook.

B. TYPES

B-1. Watertight wrist watches, covered by this specification, shall be of the following type:

Type I.

C. MATERIAL AND WORKMANSHIP

C-1. Material.-

C-1a. Movement.- The movement shall be of the best commercial material suitable for the purpose intended.

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C-1b. Dials.- The dials shall be made of metal, finished in durable lusterless black.

C-1c. Cases and backs.- The cases and backs shall be made of stainless steel.

C-1d. Other material.- Other material shall be as specified in section E.

C-2. Workmanship.- The workmanship shall be first class in every respect.

D. REQUIREMENTS COMMON TO ALL TYPES

D-1. See section E.

E. REQUIREMENTS APPLICABLE TO INDIVIDUAL TYPES

E-1. General.- Watches shall have lever escapements and shall be made to run not less than 30 hours on one winding. Watches shall be stem-winding and setting, with movements not larger than 6/0 nor smaller than 8/0 in size.

E-2. Movement.-

E-2a. Arbors, pinions, and pivots.- All train arbors, pinions, and pivots shall be made of steel, hardened, ground, and polished. The teeth of all wheels and pinions shall be cut.

E-2b. Jewels.- The movement shall have not less than 15 jewels. Train, balance, and escapement jewels shall be of the friction type. Endstone jewels shall be secured in a readily removable cap to facilitate cleaning and re-oiling.

E-2c. Balance.- The balance shall be of a nonmagnetic compensated type with a Brequet type hairspring of Elinvar, or approved equal.

E-2d. Regulator.- The movement shall have a lever regulator in its middle position, within ±10 percent of the range on each side.

E-2e. Lubrication.- The movement shall have a sufficient amount of lubricant for at least one year's operation. The lubricant shall be of such quality that the movement will function properly in temperatures of minus 20°F. to plus 160°F.

E-2f. Marking.- The number of jewels, the manufacturer's name, and the movement number shall be stamped or engraved on either the train or barrel bridge of the movement.

E-3. Dial.-

E-3a. Face finish.- The dial shall be finished in durable dull black with hour and minute markings in white. All markings shall be of durable quality.

E-3b. Graduations.- The dial shall be graduated to minutes. Each fifth minute shall be indicated by a luminous radioactive (see Fed. Spec. TT-R-58, Grade 22M) dot approximately 1/32 inch in diameter, except at the figure 12. There shall be 2 dots of a like material and size directly above the figure 12. Each tenth minute shall be marked in plain, arabic figures, except for the 60, which shall be indicated by the 2 luminous dots, one above the other.

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E-3c. Marking.- "USN BUSHIPS" shall be marked on each dial.

E-4. Hands.- The hour and minute hands shall be of the skeleton type, and shall be filled with luminous radioactive material. The second hand shall be of the sweep type and in white, except for 3/16 inch of the tip end, which shall be treated with luminous radioactive material. The luminous radioactive material shall conform to the requirements for Grade 22M of Federal Specification TT-R-58

E-5. Crystal.- The crystal shall be of tempered glass, with a metalized rim, and shall be sweated or soldered to the bezel of the case to insure a watertight joint.

E-6. Case.-

E-6a. Construction.- The case shall be of the wrist type and shall be constructed of two pieces, namely case and back. The back shall be of the wrench screw type. The case shall be provided with a screw cap on the pipe that will enclose the crown and winding arbor. The screw cap shall be provided with a gasket of Koroseal, or approved equal, to insure a watertight seal around the crown and winding arbor when the cap is screwed in place. The cap shall be attached to the case by flexible links that will permit its being removed from the pipe when winding or setting the watch. A gasket of Koroseal, or approved equal, shall be provided in the back of the case, onto which the screw back will bear when screwed in place, forming a watertight joint. The case shall have an inside snap cover and shall be provided with removable pins whereby the wrist strap may be easily removed from the case.

E-6b. Marking.- The back of the watch case shall be stamped or engraved with the following:

"USN
BUSHIPS"

E-7. Wrist strap.- Wrist straps shall be molded in accordance with Figures 1 to 6, inclusive. The material shall be a fiber forming polymeric amide resin, nontoxic and nonirritant to the skin. It shall conform to Munsell color notation N4.5 shall retain satisfactory flexibility, and shall be serviceable over a temperature range of minus 20°F. to plus 160°F.

E-8. Rates.-

E-8a. Horizontal position.- When tested as described in paragraph F-2a(1), the daily rates of watches shall not exceed 30 seconds.

E-8b. Vertical position.- When tested as described in paragraph F-2a(2), the average of daily rates shall not exceed 30 seconds. The difference between the average daily rate obtained when tested as described in paragraphs F-2a(1) and F-2a(2) shall not exceed 45 seconds.

E-8c. Pendant-down position 90°F.- When tested as described in paragraph F-2a(3), the daily rate shall not differ more than 20 seconds from the average daily rate obtained in the test described in paragraph F-2a(2).

E-8d. Pendant-down position 55°F.- When tested as described in paragraph F-2a(4), the daily rate shall not differ more than 40 seconds from the daily rate obtained in the test described in paragraph F-2a(2).

E-9. Watertightness.- There shall be not leakage of air from the watch evidenced by bubbles emanating from the back, crystal or stem when tested as described in paragraph F-2b.

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F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F-1. Watertight wrist watches shall be delivered to the Naval Observatory, Washington, D. C., for inspection and test purposes.

F-2. The watches shall be rigidly inspected by the Naval inspector to determine conformance with the requirements of this specification, and shall be subjected to the following tests:

F-2a. Rate tests.-

F-2a(1). Horizontal position - room temperature - 3 days.- Each watch shall be fully wound and allowed to run for 24 hours at room temperature. The daily rate of each watch shall be determined for a period of 3 consecutive days in a horizontal position with the dial up and with daily winding.

F-2a(2). Vertical position - room temperature - 3 days.- The daily rate of each watch shall be determined for 3 consecutive days in a vertical position at room temperature, with the pendant down and with daily winding.

F-2a(3). Pendant-down position - 90°F. - 24 hours.- Five watches, selected from each 100 or fraction thereof contracted for, shall be fully wound and subjected to a temperature of 90°F. for a period of 24 hours.

F-2a(4). Pendant-down position - 55° F. - 24 hours.- Five watches selected as specified in paragraph F-2a(3) shall be tested as described in paragraph F-2a(2), except the temperature shall be 55° F.

F-2b. Watertight test.- Each of the 5 watches selected as specified in paragraph F-2a(3) shall be suspended in a beaker of water placed in a vacuum chamber evacuated to 5 inches of mercury for a period of 3 minutes.

F-3. Rejections.-

F-3a. Inspection failures.- The individual watches failing to meet the inspection tests shall be rejected.

F-3b. Test failures.- When tests are specified on a quantity of watches that are selected as representative of a lot, and 1 or more of this number fails to meet the requirements, additional watches of the same lot will be tested immediately in order to determine the cause of the failure. If the failure is such that changes are necessary to correct the defect, the contractor shall incorporate the changes in the remaining watches on order.

F-3c. Rejected watches shall be replaced with other watches, or reworked to correct the defects, after which all specified tests shall be repeated.

F-3d. Rejected watches shall not be resubmitted for inspection without furnishing full particulars concerning previous rejections and measures taken to correct the defects.

G. PACKAGING AND PREPARATION FOR SHIPMENT

G-1. Packaging.-

G-1a. For domestic shipment.- Wrist watches shall be individually packaged and cushioned according to best commercial practice. Not more than 48 individually packaged watches shall be further packaged in a suitable style corrugated fiberboard box having a minimum Mullen test of 200 pounds.

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G-1b. For overseas shipment.- Wrist watches, packaged as described in paragraph G-1a, shall be packaged in accordance with Method IIb of Joint Army-Navy Specification JAN-P-116.

G-2. Packing.-

G-2a. For domestic shipment.- Wrist watches, packaged as described in paragraph G-1a, shall be packed with creped cellulose wadding or other suitable interior cushioning materials so applied as to maintain not less than 1-inch clearance on all sides, top and bottom, in cleated plywood boxes or nailed wood boxes conforming to the requirements of Joint Army-Navy Specifications JAN-P-105 and JAN-P-106, respectively. Excelsior, hay or straw shall not be used. No strapping will be required. In cleated plywood boxes, no water-resistant plywood will be required. Boxes shall be lined with a sealed waterproof bag, or its equivalent, made of barrier material conforming to the requirements of Joint Army-Navy Specification JAN-P-125. The seams and closures of the waterproof barrier shall be sealed with adhesive conforming to the requirements of Joint Army-Navy Specification JAN-P-140. The gross weight shall not exceed approximately 100 pounds.

G-2b. For overseas shipment.- Wrist watches, packaged as described in paragraph G-1b, shall be packed as described in paragraph G-2a, except that water resistant plywood will be required and boxes shall be strapped in accordance with the requirements of the applicable box specification.

G-2c. For domestic or overseas shipment.- Where practicable, shipping containers shall be of uniform size, and shall contain the same number of items. Containers shall be designed to fit the contents in a compact manner.

G-3. Marking.- In addition to the following and any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook:

"FRAGILE - HANDLE WITH CARE
THIS SIDE UP"

H. NOTES

H-1. Ordering information.-

H-1a. Invitations for bids and contracts or orders should specify the following:

Title, number and date of this specification.
Whether the subject commodity is to be packaged and prepared for domestic or overseas shipment (see section G).
That 100 copies, and one set of negatives for making reproductions, of a photographic parts list catalog showing part numbers, name, and number required for one watch, shall be supplied within 90 days after the first delivery on the contract or order.

H-1b. Invitations for bids should specify that bidders shall forward to the Equipment Design Section, Bureau of Ships, Navy Department, Washington 25, D. C., prior to the date fixed for the opening of bids, a sample of the watches they propose to furnish. Samples will be returned, except that the sample of the bidder awarded the contract will be retained for comparison with the watches delivered. The sample retained will apply on the first delivery. Samples should be marked with the name of the bidder, schedule number and date of opening of bids.

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H-2. Specifications and other publications - where obtainable.-

H-2a. Copies of this specification may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. When requesting specifications, refer to both title and number (or symbol).

H-2b. Copies of Navy Department specifications, Joint Army-Navy specifications, Federal specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. When requesting specifications, refer to both title and number (or symbol).

H-3. Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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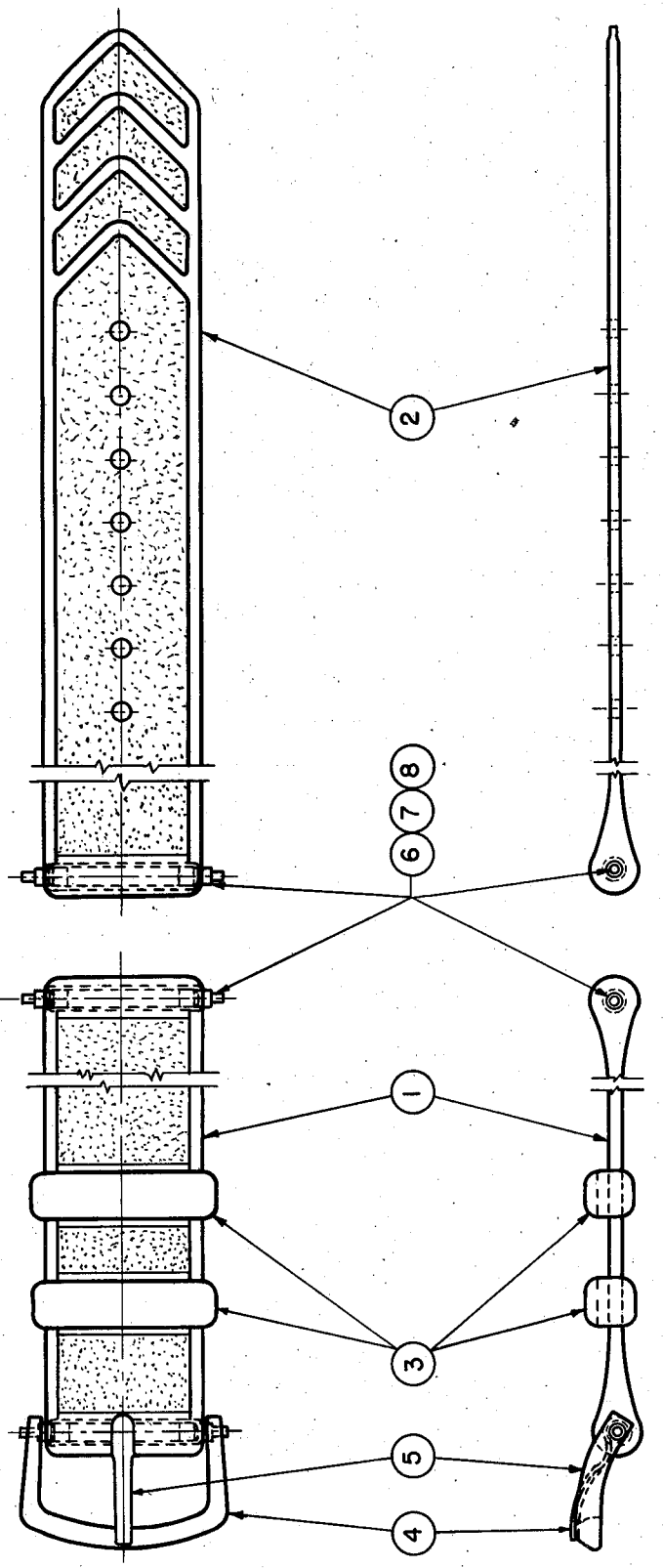


FIGURE 1 - WRIST STRAP ASSEMBLY

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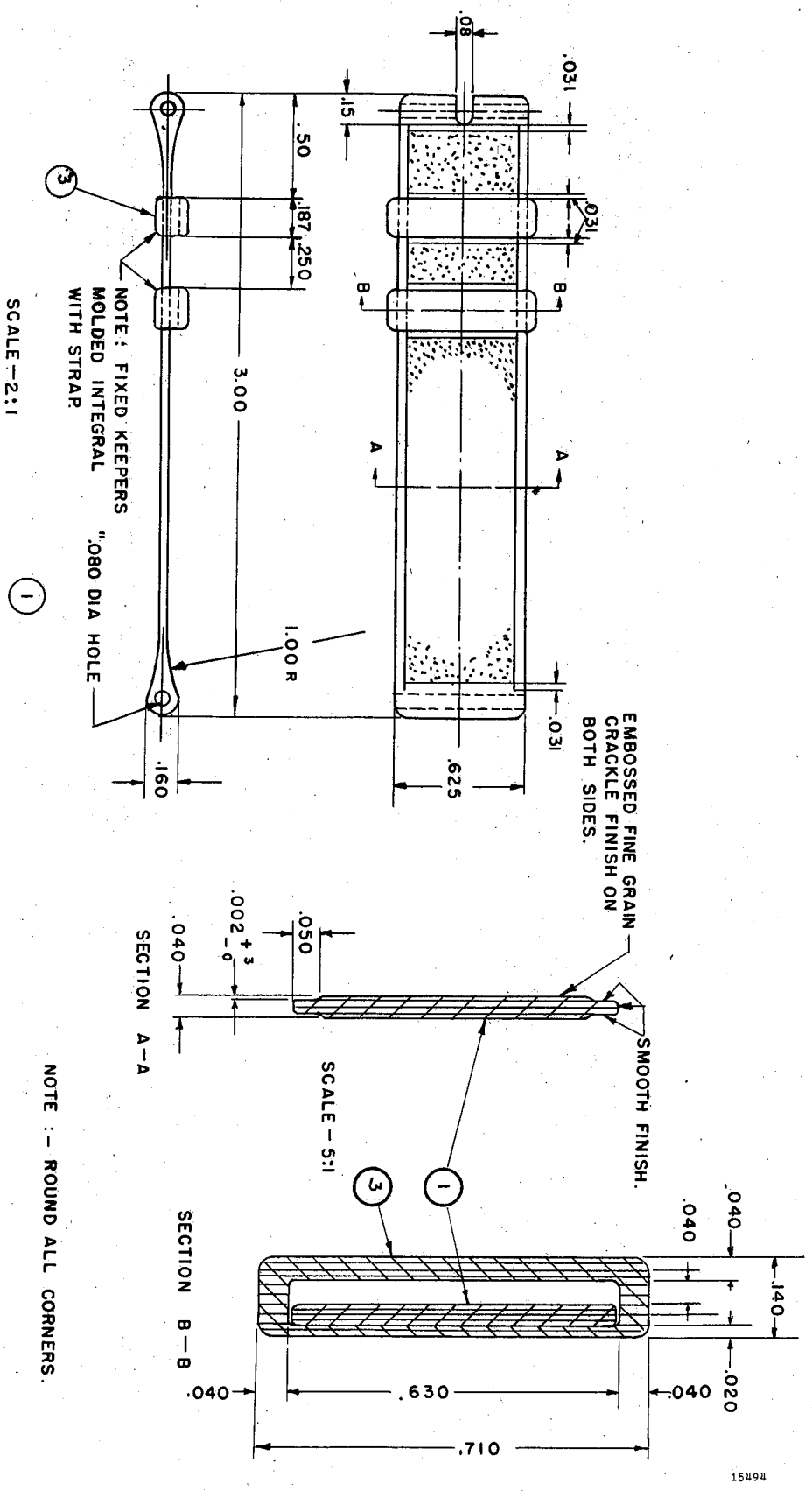


FIGURE 2. WRIST STRAP DETAILS

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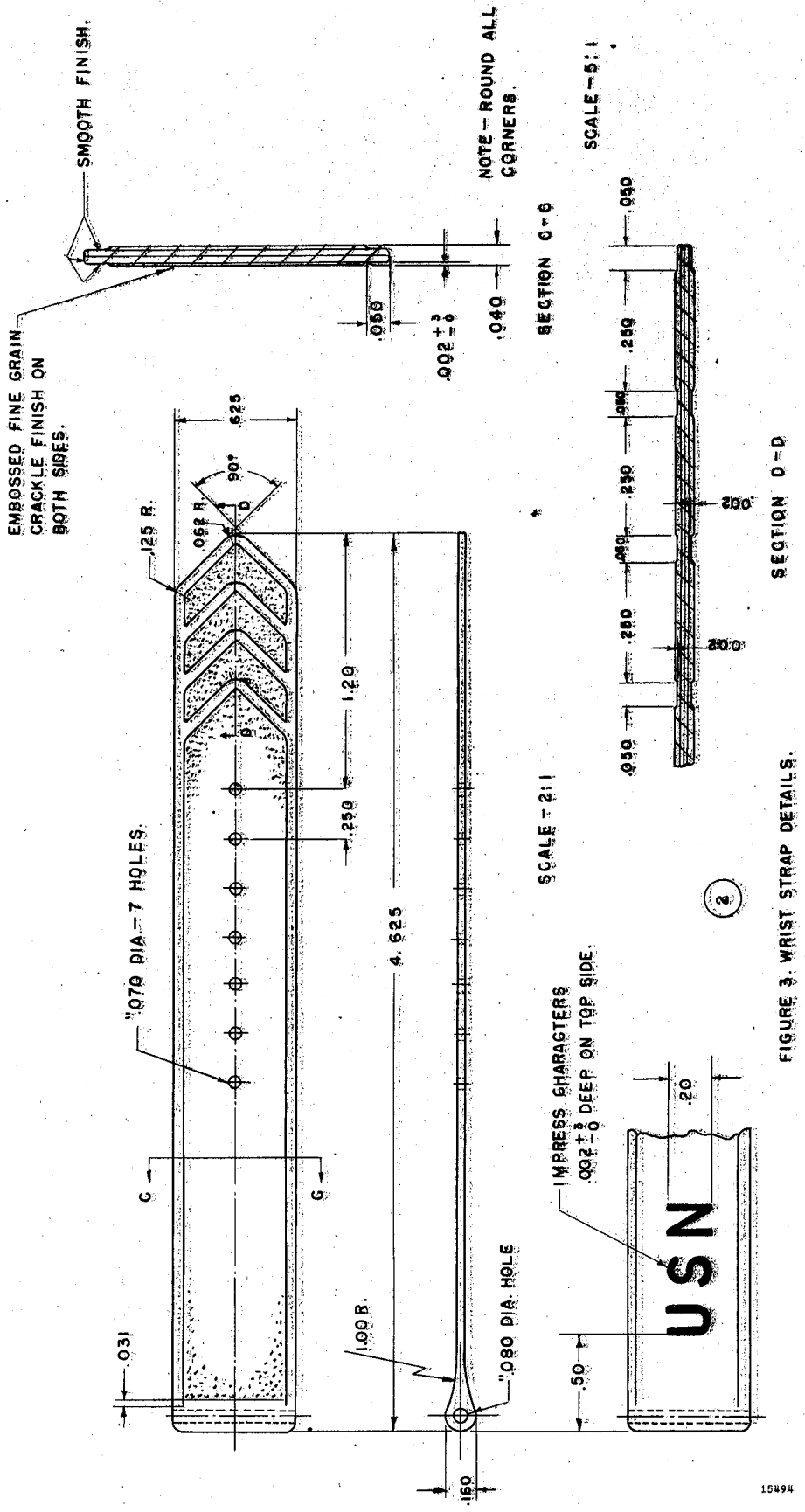
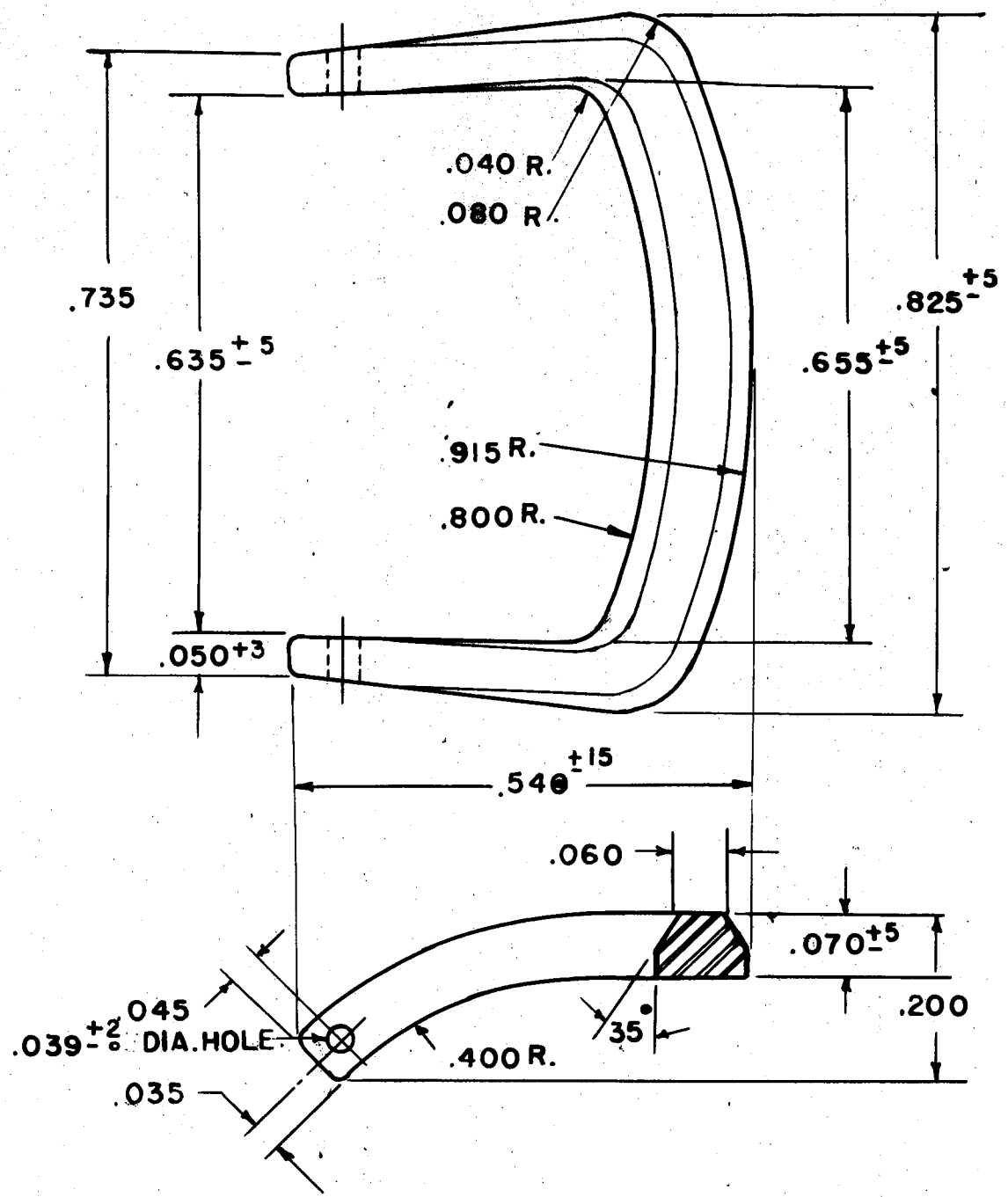


FIGURE 3. WRIST STRAP DETAILS.

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ROUND ALL CORNERS SLIGHTLY
FINISH SMOOTH & BRIGHT ALL OVER.

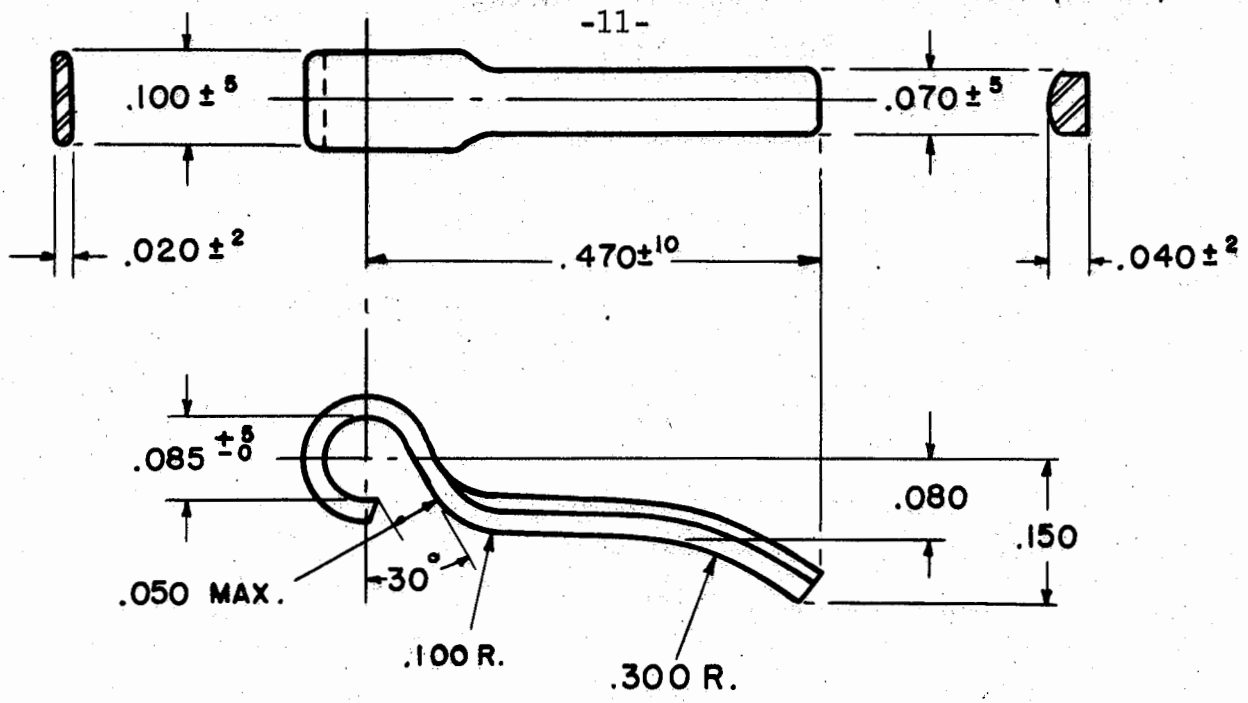
(4)

SCALE – 5:1

FIGURE 4. WRIST STRAP DETAILS.

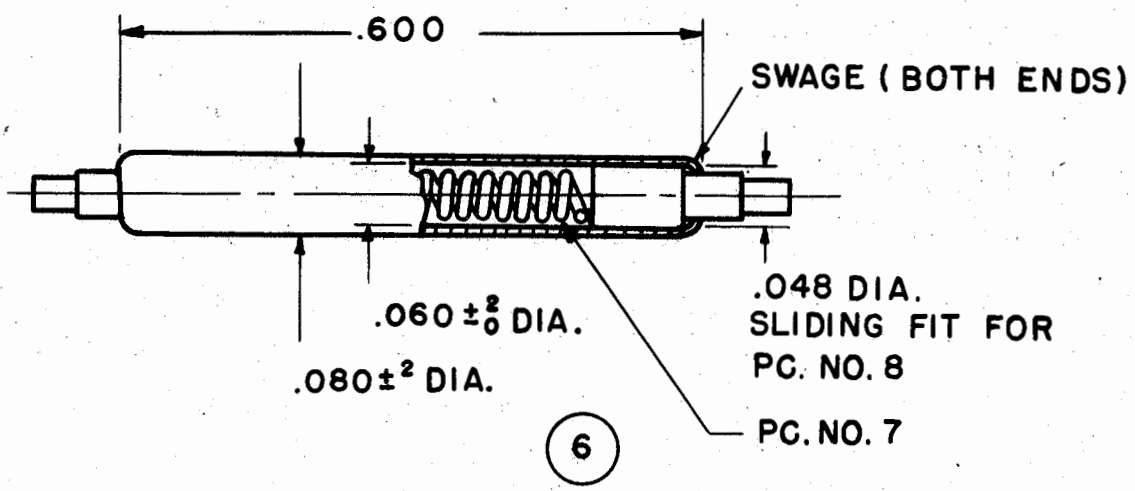
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ROUND ALL CORNERS SLIGHTLY.
FINISH SMOOTH AND BRIGHT ALL OVER.

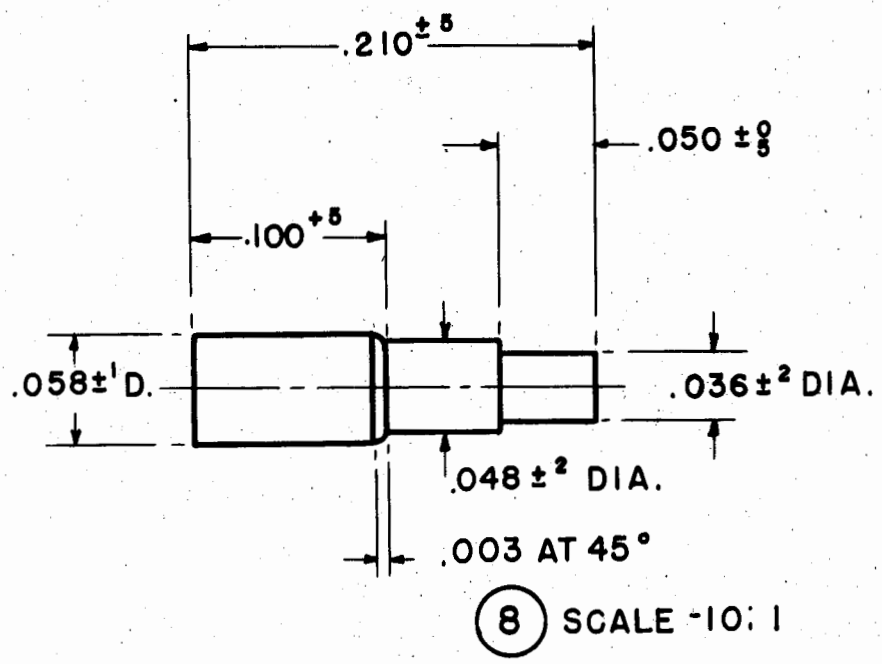
(5) SCALE - 5 : 1



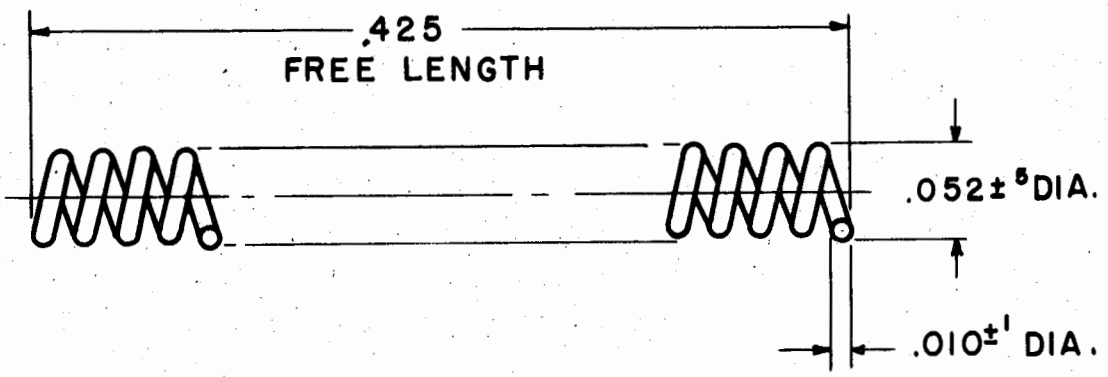
(6) SCALE - 5 : 1

FIGURE 5 . WRIST STRAP DETAILS

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(8) SCALE - 10: 1



DIRECTION OF COILING --- RIGHT OR LEFT HAND.
TOTAL NO. OF TURNS --- 20
SOLID LENGTH --- .230 MAX.

(7) SCALE - 10 : 1

FIGURE 6. WRIST STRAP DETAILS

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18W8(SHIPS)
AMENDMENT - 1
February 15, 1948

BUREAU OF SHIPS SPECIFICATION

WATCHES, WRIST (WATERTIGHT)

This amendment forms a part of Bureau of Ships Specification 18W8(SHIPS), dated January 15, 1947.

Page 3, paragraph E-9, line 1: Delete "not" and substitute "no".

Page 4, paragraph F-2b, line 2: Delete "5 inches" and substitute "3.5 inches".

Page 6, paragraph H-2a, line 3: Delete "Naval Supply Depot, Bayonne, N.J." and substitute "Naval Supply Center, Norfolk 11, Va."

Page 6, paragraph H-2b, line 4: Delete "Naval Supply Depot, Bayonne, N.J." and substitute "Naval Supply Center, Norfolk 11, Va."

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